W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

McMaster University ENGINEERING

mohawk

Capstone – Project Plan:

SUSHI ASSEMBLY LINE

PROCTECH 4TR3 – C03

September 18th, 2026

Automation Systems Engineering Technology



Prepared For:

Dr. Kostas Apostolou

Prepared By:

Nicolas Jao, 400450151

Jacob Kwan, 400453713

Jamie Williams, 400462359

McMaster University

This report contains our own work for PROCTECH – 4TR3 and has not been submitted, in whole or in part, towards fulfilling requirements in any other course, either at McMaster University or elsewhere. I/We have read McMaster's academic integrity policy and understand that plagiarism, whether intentional or unintentional, will be dealt with in accordance with university policy and practice.

Name: Nicolas Jao Date: 09/14/26 Name: Jamie Williams Date: 09/14/26

Name: Jacob Kwan Date: 09/14/26

Project Statement

Our capstone project proposes a sushi roll production machine designed to significantly reduce the time required to prepare simple maki sushi rolls. Despite being simple in structure, their manual assembly remains slow and laborious. This system automates the core steps of sushi preparation, providing each its own station along an assembly line: layering the nori and rice, adding ingredients, rolling, and slicing.



Figure 1: Image taken from https://www.justonecookbook.com/sushi-rolls/

By streamlining these tasks, this machine supports chefs by freeing them from time-consuming preparation work and enabling greater focus on higher-value culinary tasks. The design will prioritize safety features to emphasize its collaborative nature with existing chefs. It intends to increase productivity and efficiency in large-scale kitchen environments.

Project Timeline

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Jacob	Tasks	September					October				November				December
2	Jamie		Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14
3	Nicolas		8/30 - 9/05	9/06 - 9/12	9/13 - 9/19	9/20 - 9/26	9/27 - 10/03	10/04 - 10/10	10/11 - 10/17	10/18 - 10/24	10/25 - 10/31	11/01 - 11/07	11/08 - 11/14	11/15 - 11/21	11/22 - 11/28	11/29 - 12/05
4		Planning/Design														
7		Work on midterm														
8		Finish master assembly CAD								N	N					
9		Station 1: Rice Rolling														
10		Finish CAD design for motorized rice roller	N	N	N											
11		Order necessary additional parts (motor, extrusion bars)				Jamie										
12		3D print/laser cut parts		Jamie	Jamie											
13		Program motors for rice roller and platform		Jacob	Jacob	Jacob										
14		Install extrusion bars, motor, and roller					Jamie	Jamie								
15		Test station and troubleshoot						Jacob	Jacob							
16		Station 2: Placing Toppings														
17		Install reed switches				N										
18		Source end-effector for robotic arm						N	N							
19		Program UR3e cobot									Jacob	Jacob	Jacob			
20		Test station & troubleshoot											Jamie	Jamie		
21		Station 3: Rolling the Sushi														
22		Test rolling mechanism with rice and nori														
24		Program motors for rolling mechanism				Jacob	Jacob									
25		Station 4: Cutting the Roll														
26		Finish CAD design for slicers				N	N									
27		Order necessary parts (extrusion bars, motors, razors)						Jamie								
28		Program motor for slicer							Jacob	Jacob						
29		Install extrusion bars, motor, and slicers							Jamie	Jamie						
30		Test station and troubleshoot								Jacob	Jacob					
31		Final Testing														
32		Clean up wiring												N		
33		Final testing and inspections														